

Maximally Equidistributed Decimated GFSR Generators

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Abstract

This paper presents generators based on decimated sequences of Generalized Feedback Shift Register (GFSR) generators. The equivalence relation between decimated GFSR generators and Tausworthe generators are presented and a k -distributed initialization scheme is derived. We present decimated GFSR generators that are maximally equidistributed. Timing results of these generators are presented.

Key Words: linear recurring sequence, decimation, k -distribution.

1. Introduction

An M-sequence (or called PN-sequence) is defined to be a maximum-length linear recurring sequence modulo two [Golomb 1982, p.75]. Let $\langle a_t \rangle$ be an M-sequence that satisfies the following linear recurrence:

$$a_t = \sum_{i=1}^p c_i a_{t-i} \pmod{2} \quad (1)$$

and has period $T = 2^p - 1$. The polynomial

$$f(x) = 1 + \sum_{i=1}^p c_i x^i \pmod{2} \quad (2)$$

is called the characteristic polynomial of the sequence $\langle a_t \rangle$ and is primitive over Galois field GF(2).

The d -wise decimated sequence $\langle a_{dt} \rangle = \langle a_0, a_d, a_{2d}, \dots \rangle$. Sequence $\langle a_{dt} \rangle$ is also an M-sequence with the same period T , if and only if $\gcd(d, T) = 1$ [Golomb 1982, Chapter IV, Theorem 2]. All M-sequences of period T can be obtained from any given one by suitable decimation [Golomb 1982, Chapter IV, Corollary 1].

M-sequences have been widely used in the design of random number generators, such as Tausworthe generators [Tausworthe 1965] and Generalized Feedback Shift Register (GFSR) generators [Lewis and Payne 1973]. Primitive trinomials have also been widely used in the design of these random number generators; however, trinomial-based generators fail the weight distribution test [Lindholm 1968; Matsumoto and Kurita 1992;

Matsumoto and Kurita 1994].

L'Ecuyer [1996; 1999] proposes combined Tausworthe generators, which are based on products of primitive trinomials and can achieve the maximum equidistribution property, a leading selection criteria for linear generators based on recurrences modulo two. Matsumoto and Nishimura [1998] propose Mersenne Twister generators, which are based on primitive polynomials with 135 nonzero terms and guarantee the k -distribution property [Fushimi and Tezuka 1983] with any nonzero seed.

Fushimi [1990] proposes a GFSR with recursion $X_t = X_{t-3p} \oplus X_{t-3q}$. The recursion is derived from a d -wise decimation of generator with primitive trinomial $x^p + x^q + 1$, $3d \equiv 1 \pmod{T}$. Ferrenberg et al. [1992] use every fifth call of generator R250 [Kirkpatrick and Stoll 1981], which is based on primitive trinomial $x^{250} + x^{103} + 1$. Ziff [1998] shows that 3-wise, 5-wise, and 7-wise decimation of generators with primitive trinomials actually are generators with primitive pentanomials.

This paper presents generators based on decimated sequences of GFSR generators. The equivalence relation between decimated GFSR generators and Tausworthe generators are presented and a k -distributed initialization scheme is thus derived, where $1 \leq k \leq \lfloor p/\ell \rfloor$, and ℓ is an integral power of 2. We search the d -wise decimation of GFSR generators with primitive pentanomials, for d in 3 to 99. We find several decimated GFSR generators that are maximally equidistributed. Timing results of these generators are presented.

2. The Equivalence Relations

We follow the definitions in Fushimi [1989] and Golomb [1982]. Let

$$R = \{ r \mid 1 \leq r < T, \gcd(r, T) = 1 \}. \quad (3)$$

R forms a group under multiplication modulo T . We denote by σ^{-1} the inverse of σ in the multiplicative group. Let coset

$$C_1 = \{1, 2, 2^2, 2^3, 2^4, \dots, 2^{p-1}\} \quad (4)$$

be the multiplier subgroup. Taking any element from R and multiplying it by each number in C_1 obtain the other cosets, denoted as $C_2, \dots, C_K$, where $K = \phi(T)/p$, $|R| = \phi(T)$, and $\phi(T)$ is Euler's totient function.

We denote by $f_1(D)$ the primitive polynomial associated with $\langle a_t \rangle$ and by $f_i(D)$ one associated with $\langle a_{dt} \rangle$, $d \in C_i$, $2 \leq i \leq K$.

Using the M-sequence $\langle a_t(f) \rangle$, we construct three sequences of ℓ -bit ($2 \leq \ell \leq p$) binary integers as follows.

(1) Tausworthe sequence $\langle X_t(f; \sigma) \rangle$:

$$X_t = a_{\sigma} a_{\sigma+1} a_{\sigma+2} \dots a_{\sigma+\ell-1}, \sigma \in R \quad (5)$$

(2) GFSR sequence $\langle Y_t(f; \tau) \rangle$:

$$Y_t = a_t a_{t+\tau} a_{t+2\tau} \dots a_{t+(\ell-1)\tau}, \tau \in R \quad (6)$$

(3) Decimated GFSR sequence $\langle Z_t(f; \tau; d) \rangle$, a

d -wise decimation of $\langle Y_t(f; \tau) \rangle$:

$$Z_t = Y_{dt} = a_{dt} a_{dt+\tau} a_{dt+2\tau} \dots a_{dt+(\ell-1)\tau}, \tau \in R \quad (7)$$

Fushimi [1989, Theorem] shows that for any p and ℓ , the following equivalence relations hold.

$$\langle X_t(f_1; \sigma) \rangle \cong \langle Y_t(f_i; \sigma^{-1}) \rangle \text{ if } \sigma \in C_i \quad (8)$$

$$\langle Y_t(f_1; \tau) \rangle \cong \langle X_t(f_j; \tau^{-1}) \rangle \text{ if } \tau \in C_j \quad (9)$$

The following theorem shows two more equivalence relations.

THEOREM. For any p and ℓ , the following equivalence relations hold.

$$\langle Z_t(f_1; \tau; d) \rangle \cong \langle X_t(f_u; \tau^{-1}d) \rangle \text{ if } \tau \in C_u \quad (10)$$

$$\langle Z_t(f_1; \tau; d) \rangle \cong \langle Y_t(f_v; \tau d^{-1}) \rangle \text{ if } d \in C_v \quad (11)$$

PROOF: For any integer m , $0 \leq m \leq \ell - 1$, we consider the sequence $\langle a_{dt+m\tau}(f_1) \rangle$

$$\langle a_{dt+m\tau}(f_1) \rangle \cong \langle a_{\tau(\tau^{-1}dt+m)}(f_1) \rangle \cong \langle a_{\tau^{-1}dt+m}(f_u) \rangle \text{ if } \tau \in C_u \quad (12)$$

Since $a_{dt+m\tau}(f_1)$ constitutes the $(m+1)$ -st (from the left) bit of $\langle Z_t(f_1; \tau; d) \rangle$ and $a_{\tau^{-1}dt+m}(f_u)$ constitutes the $(m+1)$ -st bit of $\langle X_t(f_u; \tau^{-1}d) \rangle$, the above relation proves the equivalence relation between a decimated GFSR sequence and a Tausworthe sequence. On the other hand, for any integer m , $0 \leq m \leq \ell - 1$, we also consider the sequence $\langle a_{dt+m\tau}(f_1) \rangle$

$$\langle a_{dt+m\tau}(f_1) \rangle \cong \langle a_{d(t+m\tau d^{-1})}(f_1) \rangle \cong \langle a_{t+m\tau d^{-1}}(f_v) \rangle \text{ if } d \in C_v \quad (13)$$

Similarly, we obtain the equivalence relation between a

decimated GFSR sequence and a GFSR sequence.

Q.E.D.

Let ℓ be an integral power of 2 and $\tau = d/\ell$.

Then, $\tau^{-1}d = \ell$ and $d \in C_u$. The following equivalence relation holds:

$$\langle Z_t(f_1; \tau; d) \rangle \cong \langle X_t(f_u; \ell) \rangle \text{ if } d \in C_u \quad (14)$$

We can thus use $\langle X_t(f_u; \ell) \rangle$, $d \in C_u$, $0 \leq t < p$, as initial values for the sequence $\langle Z_t(f_1; \tau; d) \rangle$. This initialization is k -distributed for $1 \leq k \leq \lfloor p/\ell \rfloor$ based on the theorems by Tausworthe [1965] and Fushimi [1989].

3. Searching Maximally Equidistributed Generators

We search decimated GFSR generators that are maximally equidistributed [L'Ecuyer 1996]. The GFSR generators used are with primitive pentanomials $x^p + x^q + x^r + x^s + 1$ selected from Kurita and Matsumoto [1991], where p is a Mersenne prime. These generators are labeled as K1 to K6 as shown in Table 1. Since the period $T = 2^p - 1$ is also a prime, and $\gcd(d, T) = 1$, any d -wise decimation of these GFSR generators have the same period T . We search the d -wise decimation of these GFSR generators for d in 3 to 99, and d being odd. For d being even, d -wise decimation of these GFSR generators is equivalent to the $d/2$ -wise decimation of the generator.

We find seven decimated GFSR generators that are maximally equidistributed as shown in Table 1: $d \in \{31, 65\}$ for generator K1; $d \in \{7, 15, 53\}$ for K2; $d = 61$ for K4; and $d = 81$ for K5. We find none of decimated sequences of K3 and K6 are maximally equidistributed for d searched.

Table 1 also lists the number of nonzero terms (n) of these decimated GFSR generators and the ratio n/p . We make a comparison with the following generators: taus88, a three-component combined Tausworthe generator [L'Ecuyer 1996]; TT800, a tempered twisted GFSR generator [Matsumoto and Kurita 1994]; and MT19937, a Mersenne twister generator [Matsumoto and Nishimura 1998]. Six of the seven decimated GFSR generators have the ratio n/p greater than 0.3. In the three compared generators, only taus88 has this ratio close to 0.3. The ratio of MT19937 is 0.0068.

The right-most column of Table 1 shows the timing results of these generators. Function `get_rand`, which produces a 32-bit random integer, was executed 100,000,000 times. All programs were executed five times and the averages were taken. The test platform is a PC with Pentium4 2.0 GHz and 256 MB memory. The compiler used is DJGPP 2.03 with flag `-O`. These

decimated GFSR generators are implemented by using a power-of-2 buffer [Ziff 1998, p.392]. All these decimated GFSR generators run much slower than taus88, TT800, and MT19937. The Appendix is the sample C code for decimated GFSR generator with $d = 81$ and GFSR generator K5.

4. Conclusions

We have presented the equivalence relation between decimated GFSR generators and Tausworthe generators and have derived a k -distributed initialization scheme based on this relation. We have also found seven maximally equidistributed decimated GFSR generators with primitive pentanomials. Six of the seven decimated GFSR generators have the ratio n/p greater than 0.3. All these decimated GFSR generators run much slower than taus88, TT800, and MT19937.

Appendix. The C Program for Decimated GFSR generators

The C preprocessor constants D , P , Q , R , and S are defined for a d -wise decimated generator with primitive polynomial $x^p + x^q + x^r + x^s + 1$, $p > q > r > s$. LOG_W is defined as the number of bits in a machine word. Function set_seed uses generator SR(28, 3, 31) [Anderson 1990] (macro rand) to generate p random bits for the initialization of the equivalent Tausworthe sequence $\langle X_t(f_u; \ell) \rangle$, $d \in C_u$, which state is stored in a bit buffer (unsigned char init_buf[P]). Function set_seed then generates the first p words of the equivalent Tausworthe sequence for the initialization of the decimated GFSR generator. Variable IX is the index to the buffer (unsigned int buffer[POW2]). POW2 is the smallest power of 2 greater than p ; M is defined as $(\text{POW2} - 1)$. Bit-wise “and” with M is effective to cause the random numbers cycle around the buffer.

```
#ifndef D
#   define D      81
#endif
#ifndef P
#   define P      1279
#   define Q      988
#   define R      630
#   define S      339
#   define POW2   2048
#endif
#define LOG_W     32
#define M         (POW2-1)
#define rand(seed) \
```

```
(tmp = seed ^ (seed >> 3), \
seed = (tmp ^ (tmp << 28)) & 0x7ffffff)
static unsigned char init_buf[P];
static unsigned int buffer[POW2];
static int IX;
void set_seed(unsigned int seed)
{
    int i, j, k;
    unsigned int tmp, x;
    for(i=0; i<P; ) { /* init P bits */
        x = rand(seed);
        for(j=0; j<31 && i+j<P; j++)
            init_buf[i+j] = (x >> j) & 1;
        i = i + j;
    }
    IX = 0;
    for(i=0; i<P; i++) /* init P words */
    {
        unsigned int w = 0;
        unsigned char b, t;
        int ixq;
        for(j=0; j<LOG_W; j++) {
            for(k=0; k<D; k++) {
                t = init_buf[IX];
                if ((ixq=IX-Q) < 0) ixq += P;
                t ^= init_buf[ixq];
                if ((ixq=IX-R) < 0) ixq += P;
                t ^= init_buf[ixq];
                if ((ixq=IX-S) < 0) ixq += P;
                t ^= init_buf[ixq];
                b = init_buf[IX] = t;
                if (++IX == P) { IX=0; }
            }
            w = (w<<1) | (b&1);
        }
        buffer[i] = w;
    }
    IX = P;
}

unsigned int get_rand()
{
    int i;
    unsigned int tmp;
    for(i=0; i<D; i++)
    {
        tmp = buffer[IX&M] = buffer[(IX-P)&M] ^
buffer[(IX-Q)&M] ^ buffer[(IX-R)&M] ^
buffer[(IX-S)&M];
        IX++;
    }
    return tmp;
}
```

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Table 1. Maximally equidistributed decimated GFSR generators.

Generator	(p, q, r, s) or p	d	# nonzero terms (n)	n / p	execution time
K1	(89, 69, 40, 20)	31	29	0.3258	24.58 s.
		65	37	0.4157	46.98 s.
K2	(127, 83, 63, 22)	7	25	0.1969	7.93 s.
		15	55	0.4331	17.26 s.
		53	52	0.4094	42.22 s.
K3	(521, 447, 197, 86)	none	-	-	-
K4	(607, 461, 307, 167)	61	249	0.4102	44.25 s.
K5	(1279, 988, 630, 339)	81	459	0.3589	66.53 s.
K6	(3217, 2381, 1621, 809)	none	-	-	-
taus88	88	n.a.	26	0.2955	1.32 s.
TT800	800	n.a.	93	0.1163	2.36 s.
MT19937	19937	n.a.	135	0.0068	2.03 s.

n.a. = not applicable